

WEATHER FORECASTING USING DATA MINING AND PREDICTIVE ANALYTICS MODELS

A Project Report Submitted for the Course: Data Mining

Abstract

Weather forecasting is one of the most significant applications of data mining and predictive analytics because weather conditions affect many sectors including agriculture, transportation, tourism, and disaster management. Traditional weather forecasting methods often rely on statistical and manual techniques, which may not always provide highly accurate predictions. With the advancement of artificial intelligence and deep learning, weather forecasting systems have become more efficient and intelligent.

This project presents a **Weather Forecasting Using Data Mining and Predictive Analytics Models** system developed using deep learning techniques. The project aims to predict weather conditions and forecast temperature using machine learning and neural network-based models. Two predictive models are implemented in this system: an **Artificial Neural Network (ANN)** model for weather classification and a **Long Short-Term Memory (LSTM)** model for temperature forecasting.

The dataset used for this project contains historical weather information including precipitation, temperature, and wind speed. Data preprocessing techniques such as normalization and feature scaling were applied to improve model performance. The developed system was deployed through a **Streamlit-based web application**, allowing users to interact with the forecasting system in real time.

The results show that deep learning techniques can effectively analyze historical weather data and generate accurate predictions. This project demonstrates the practical implementation of data mining concepts in predictive weather analysis and provides a foundation for future improvements in intelligent forecasting systems.

Keywords: Data Mining, Weather Forecasting, Deep Learning, Artificial Neural Network, LSTM, Predictive Analytics, Streamlit.

1. Introduction

Weather forecasting is an essential component of modern society as it helps people and organizations make informed decisions. Accurate weather prediction assists in agricultural planning, disaster prevention, transportation scheduling, and tourism management. Weather information is especially important in countries where climate changes significantly influence economic activities.

In recent years, large volumes of weather-related data have become available through digital systems and meteorological organizations. This increase in data availability has created opportunities to use **data mining techniques** for discovering hidden patterns and trends in weather datasets. Data mining allows researchers to extract meaningful information from historical records and use predictive algorithms for forecasting future outcomes.

Traditional weather prediction approaches often depend on numerical and statistical models that require extensive calculations and domain expertise. However, modern machine learning and deep learning methods provide automated solutions capable of learning complex weather patterns from historical data.

This project introduces a weather forecasting system developed using predictive analytics and deep learning techniques. The system combines an **Artificial Neural Network (ANN)** for predicting weather conditions and an **LSTM (Long Short-Term Memory)** network for forecasting future temperatures. Furthermore, the project includes a user-friendly web interface developed using Streamlit to make predictions accessible to users.

The primary objective of this project is to demonstrate how data mining techniques can be applied to solve real-world forecasting problems efficiently and accurately.

2. Problem Statement

Weather conditions are dynamic and difficult to predict because they depend on multiple environmental factors such as precipitation, temperature, humidity, and wind speed. Traditional prediction systems may not always adapt effectively to changing environmental patterns.

Many forecasting systems also require expert-level interpretation and may not provide real-time accessibility for ordinary users. Therefore, there is a need for an intelligent system capable of:

- Predicting weather conditions automatically.
- Forecasting future temperature values.
- Providing users with a simple and interactive interface.
- Improving prediction accuracy using historical weather data.

This project addresses these challenges by implementing deep learning-based forecasting models for weather analysis.

3. Objectives of the Project

The main objectives of this project are:

1. To develop a weather forecasting system using data mining techniques.
2. To classify weather conditions using an Artificial Neural Network (ANN).
3. To forecast future temperature using an LSTM model.

4. To preprocess weather data for improved prediction performance.
 5. To build an interactive web application using Streamlit.
 6. To evaluate the performance of predictive analytics models.
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4. Literature Review

Weather forecasting has been an important area of research for many years. Researchers have used various statistical and machine learning approaches to improve forecasting accuracy.

Traditional forecasting techniques relied heavily on statistical methods and historical averages. However, these methods often struggled to capture nonlinear relationships among weather variables.

Machine learning introduced new possibilities for predictive weather analysis. Artificial Neural Networks (ANNs) have been widely used because they can learn hidden relationships between variables and produce better predictions than traditional methods.

Deep learning methods such as **Recurrent Neural Networks (RNN)** and **Long Short-Term Memory (LSTM)** models became increasingly popular for time-series forecasting. LSTM networks are especially useful because they can retain long-term dependencies in sequential weather data.

Recent studies have shown that combining multiple deep learning approaches can improve forecasting performance. Inspired by these developments, this project uses both ANN and LSTM techniques to classify weather conditions and forecast temperature.

5. Methodology

The methodology of this project consists of several important stages including data collection, preprocessing, model development, training, evaluation, and deployment.

5.1 Data Collection

The dataset used in this project is the **Seattle Weather Dataset**, which contains historical weather observations. The dataset includes weather-related features such as:

- Precipitation
- Maximum Temperature
- Minimum Temperature
- Wind Speed
- Weather Condition

These attributes are used for prediction and classification purposes.

5.2 Data Preprocessing

Data preprocessing is an important step in data mining because raw data may contain inconsistencies and variations.

The following preprocessing techniques were applied:

1. Feature Selection

The important weather variables selected for prediction include:

- Precipitation
- Maximum Temperature
- Minimum Temperature
- Wind Speed

2. Data Scaling

The project uses **MinMaxScaler** for feature normalization. Scaling transforms numerical values into a standard range, improving model efficiency and reducing prediction errors.

3. Label Encoding

Weather labels were converted into numerical values using **Label Encoder** to make them compatible with neural network training.

4. Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

6. System Design and Architecture

The system architecture consists of three main components:

1. Dataset Processing Module

This module prepares historical weather data for training and prediction.

2. Predictive Analytics Models

Two deep learning models were implemented:

Artificial Neural Network (ANN)

Used for weather classification.

Long Short-Term Memory (LSTM)

Used for forecasting future temperatures.

3. User Interface

A Streamlit web application provides a user-friendly interface for weather prediction and temperature forecasting.

7. Model Development

7.1 Weather Classification Model (ANN)

An **Artificial Neural Network (ANN)** was developed to classify weather conditions based on user input.

Input Features:

- Precipitation
- Maximum Temperature
- Minimum Temperature
- Wind Speed

Architecture:

The ANN model contains:

- Input Layer
- Dense Hidden Layer (64 neurons)
- Dropout Layer (0.3)
- Dense Hidden Layer (32 neurons)
- Dropout Layer (0.2)
- Output Layer using Softmax Activation

The model was compiled using:

- **Optimizer:** Adam
- **Loss Function:** Categorical Crossentropy
- **Metric:** Accuracy

The ANN model predicts weather categories such as rain, sunny, fog, and cloudy weather.

7.2 Temperature Forecasting Model (LSTM)

A **Long Short-Term Memory (LSTM)** network was developed for forecasting next-day temperature.

The model learns patterns from previous temperature sequences and predicts future maximum and minimum temperatures.

Model Architecture:

The LSTM model includes:

- LSTM Layer (64 Units)
- Dropout Layer
- LSTM Layer (32 Units)
- Dense Layer
- Output Layer

Sequence Length:

The model uses weather data from the previous **7 days** to forecast future temperature.

Loss Function:

Mean Squared Error (MSE)

Evaluation Metric:

Mean Absolute Error (MAE)

This approach helps the model identify seasonal and sequential temperature patterns.

8. Tools and Technologies Used

Several technologies were used in this project:

Technology	Purpose
Python	Programming Language
Pandas	Data Handling
NumPy	Numerical Operations
Scikit-Learn	Data Preprocessing
TensorFlow/Keras	Deep Learning Models
Streamlit	Web Application Development
Matplotlib	Data Visualization

These technologies enabled efficient implementation of predictive analytics and model deployment.

9. Implementation

The implementation phase involved training predictive models and integrating them into a web application.

The Streamlit application provides three main sections:

Home Page

Displays:

- Dataset overview
- Statistical summary
- Weather distribution charts
- Temperature trends

Weather Prediction Page

Allows users to enter:

- Precipitation
- Maximum Temperature
- Minimum Temperature
- Wind Speed

The ANN model predicts the likely weather condition and confidence percentage.

Temperature Forecasting Page

Users input weather information for the previous **7 days**, and the LSTM model predicts:

- Maximum temperature
- Minimum temperature

This interactive interface improves usability and accessibility.

10. Results and Discussion

The developed system successfully predicts weather conditions and forecasts temperature using historical data.

The ANN classification model achieved satisfactory accuracy in identifying weather categories based on weather variables.

Similarly, the LSTM model demonstrated good performance in temperature prediction by learning sequential weather trends.

The use of feature scaling significantly improved model efficiency and reduced prediction error.

The Streamlit application successfully integrated the trained models and allowed users to interact with the forecasting system in real time.

Overall, the results indicate that deep learning models can effectively support weather prediction tasks using historical datasets.

11. Advantages and Limitations

Advantages

- Provides automated weather prediction.
- Uses deep learning for better accuracy.
- Interactive and user-friendly interface.
- Real-time predictions.
- Efficient use of historical weather data.

Limitations

- Dataset is limited to Seattle weather data.
 - Prediction accuracy depends on data quality.
 - External climate factors are not included.
 - Limited geographical adaptability.
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12. Future Enhancements

The project can be improved in the following ways:

1. Use real-time weather APIs.
 2. Include additional weather variables such as humidity and pressure.
 3. Train models using larger datasets.
 4. Add graphical dashboards for better visualization.
 5. Deploy the system online for public use.
 6. Improve model accuracy using advanced architectures.
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13. Conclusion

This project successfully demonstrated the use of **data mining and predictive analytics** techniques in weather forecasting. Deep learning models including ANN and LSTM were implemented to classify weather conditions and predict future temperature values.

The results indicate that historical weather data can be effectively used for intelligent forecasting. Data preprocessing, feature scaling, and sequence learning significantly improved prediction quality.

The Streamlit application made the system practical and user-friendly by enabling real-time interaction. Overall, this project highlights the importance of data mining in solving real-world prediction problems and provides a strong foundation for future intelligent weather forecasting systems.

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